Adnan Shahriar, PhD

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PROFESSIONAL SUMMARY

Ph.D. mechanical engineer combining HVAC product development, CFD/FEA, scientific programming, and prototype testing. At Rheem, developed and experimentally evaluated fan, air-handling, and condensate-management designs, including a fan-shroud prototype demonstrating more than 80% higher airflow efficiency (CFM/W) than the previous design. Additional experience includes real-time C/C++ HVAC plant modeling and coupled thermal-fluid simulation.

TECHNICAL SKILLS

Thermal and Fluids: HVAC airflow systems, Heat transfer, CFD, Transient thermal modeling, Convective thermal-fluid coupling

Mechanical Design and Analysis: FEA, Structural optimization, Parametric CAD, Design for manufacturability, Assembly and clearance analysis

Prototypes and Validation: 3D printing, Airflow testing, Variable-geometry test fixtures, Experimental modal testing, Model calibration

Programming and CAD: Python, C/C++, CadQuery, STEP geometry generation, Custom numerical solvers, Engineering workflow automation

PROFESSIONAL EXPERIENCE

Project Engineer — Rheem Manufacturing (Friedrich Air Conditioning) | San Antonio, TX *Mar 2025–Present*

- **Outdoor Fan-Shroud Optimization**
  - Used CFD to optimize shroud geometry and improve airflow distribution across the condenser coil. Fabricated and tested a prototype, demonstrating more than 80% improvement in airflow efficiency (CFM/W) relative to the previous design.
- **Airflow-Induced Condensate Management Invention**
  - Invented a passive condensate entrainment and slinging system that uses fan-generated pressure differences to draw condensed water into the fan airflow path and improve condensate slinging performance.
  - Performed CFD simulations to investigate airflow behavior, pressure distribution, and condensate entrainment mechanisms.
  - Designed, fabricated, and iteratively refined 3D-printed prototypes to demonstrate the mechanism.
  - Conducted experimental testing to validate condensate entrainment, slinging effectiveness, and overall system performance.
- **VPAK Air-Handling Unit Development**
  - Designed an air-handling unit that delivered approximately 25% higher airflow than the existing configuration. Fabricated and tested 3D-printed prototypes to assess airflow and refine design assumptions.
- **Reconfigurable Fan Test Fixture**
  - Designed and fabricated a modular, 3D-printed cross-flow fan test fixture with adjustable tongue and scroll geometry and clearances, enabling multiple configurations to be tested without rebuilding the full housing.
- **Multi-Material Product FEA Analysis**
  - Performed finite element analysis (FEA) to support product designs using ABS and ASA plastics, aluminum and steel sheet metal, and expanded polypropylene (EPP) at different densities.
- **Structural FEA and Assembly Optimization of Modular AHU**
  - Performed comprehensive finite element analysis (FEA) of the nine-segment modular AHU assembly to evaluate structural integrity, press-fit connections, and load-bearing capacity. Analyzed deformation under fan and motor loads to maintain critical clearances between the rotating fan lip and stationary bellmouth, preventing mechanical interference and supporting reliable prototype operation.
- **Invention Disclosures**
  - Submitted four patent disclosures covering HVAC innovations, including fan shroud optimization, condensate management, and experimental testing systems.

- **Modular VPAK Air-Handling Unit Prototype**
  - Designed and developed a nine-segment modular VPAK air-handling unit (AHU) prototype, allowing individual sections to be modified independently for airflow optimization and rapid accommodation of design changes, including heater, coil, and component configurations. Used low-cost 3D printing to accelerate design iteration while maintaining a unified, single-piece AHU configuration for production.
- **Parametric Fan-Shroud Generation**
  - Developed a Python/CadQuery-based tool that converts fan diameter, coil geometry, component locations, and design clearances into 3D shroud geometry and STEP files, reducing repetitive CAD modeling and accelerating design iteration across product configurations.
- **Motor-Mount Structural Optimization**
  - Redesigned a motor-mount structure using FEA, reducing material usage by more than 50% while maintaining stiffness and vibration requirements. Simplified injection-molded components to reduce secondary manufacturing operations and assembly complexity.
- **Cross-Flow Fan Aerodynamics**
  - Used CFD to investigate tongue-vortex interaction, recirculation, eccentric vortex behavior, and the sensitivity of fan performance to scroll and housing geometry.
- **Fan-Induced Vibration and Modal Analysis**
  - Performed FEA-based modal analysis to identify natural frequencies of product components and assess resonance risk by comparing them with fan rotational frequency (RPM/60) and blade-passing frequency.

Senior Simulation Integration Engineer — L&T Technology Services | Plano, TX *Jan 2024–Mar 2025*

- **Real-Time C/C++ HVAC Plant Model for SiL and HiL**
  - Developed a real-time C/C++ executable HVAC plant model for Software-in-the-Loop (SiL) and Hardware-in-the-Loop (HiL) applications, combining finite element solution of transient heat transfer and velocity-pressure-coupled Navier-Stokes equations with lumped-capacitance heater/cooler models, convective thermal-fluid coupling, and a proportional-derivative (PD) HVAC controller model.
- **Modular GUI for Aerodynamic Data Analysis**
  - Developed a modular GUI for aerodynamic data analysis with well-defined input/output interfaces, improving maintainability and reducing computational and memory overhead.

Graduate Researcher — Resilient Extraterrestrial Habitats Institute | San Antonio, TX *Dec 2019–Jan 2024*

- **Custom 3D Multiphysics Finite Element Solvers**
  - Developed custom three-dimensional finite element solvers from first principles for structural mechanics, transient heat transfer, and fluid dynamics, including coupled structural-thermal habitat models and interior thermal-fluid simulations.
- **NASA Modular-Coupled Virtual Testbed (MCVT)**
  - Developed and integrated structural, thermal, and fluid plant models for NASA's Modular-Coupled Virtual Testbed (MCVT), a system-of-systems simulation environment for virtual testing of resilient extraterrestrial habitats.
- **Thermal Protective Layer Topology Optimization**
  - Investigated thermo-mechanical topology optimization of lunar habitat thermal protective layers under different environmental conditions and lunar latitudes.
- **Experimental Structural Digital Twin and Model Updating**
  - Created a structural digital twin of an experimental habitat by acquiring impact-hammer and accelerometer data, identifying natural frequencies and mode shapes, using Crypton optical geometry measurements, and calibrating a parameterized beam-element model to match identified modal properties and measured acceleration histories.

SELECTED TECHNICAL DEMONSTRATIONS

HVAC duct CFD: https://www.linkedin.com/feed/update/urn:li:activity:7258219618144538624/

Lunar habitat thermal topology optimization: https://youtu.be/EGVZgg1pb6Q

MCVT structural-mechanical model: https://youtu.be/TAA66P_UXIM

EDUCATION

Ph.D., Mechanical Engineering — University of Texas at San Antonio, 2023 | GPA: 3.94

B.S., Aeronautical Engineering (Aerospace Major) — Military Institute of Science and Technology, 2015